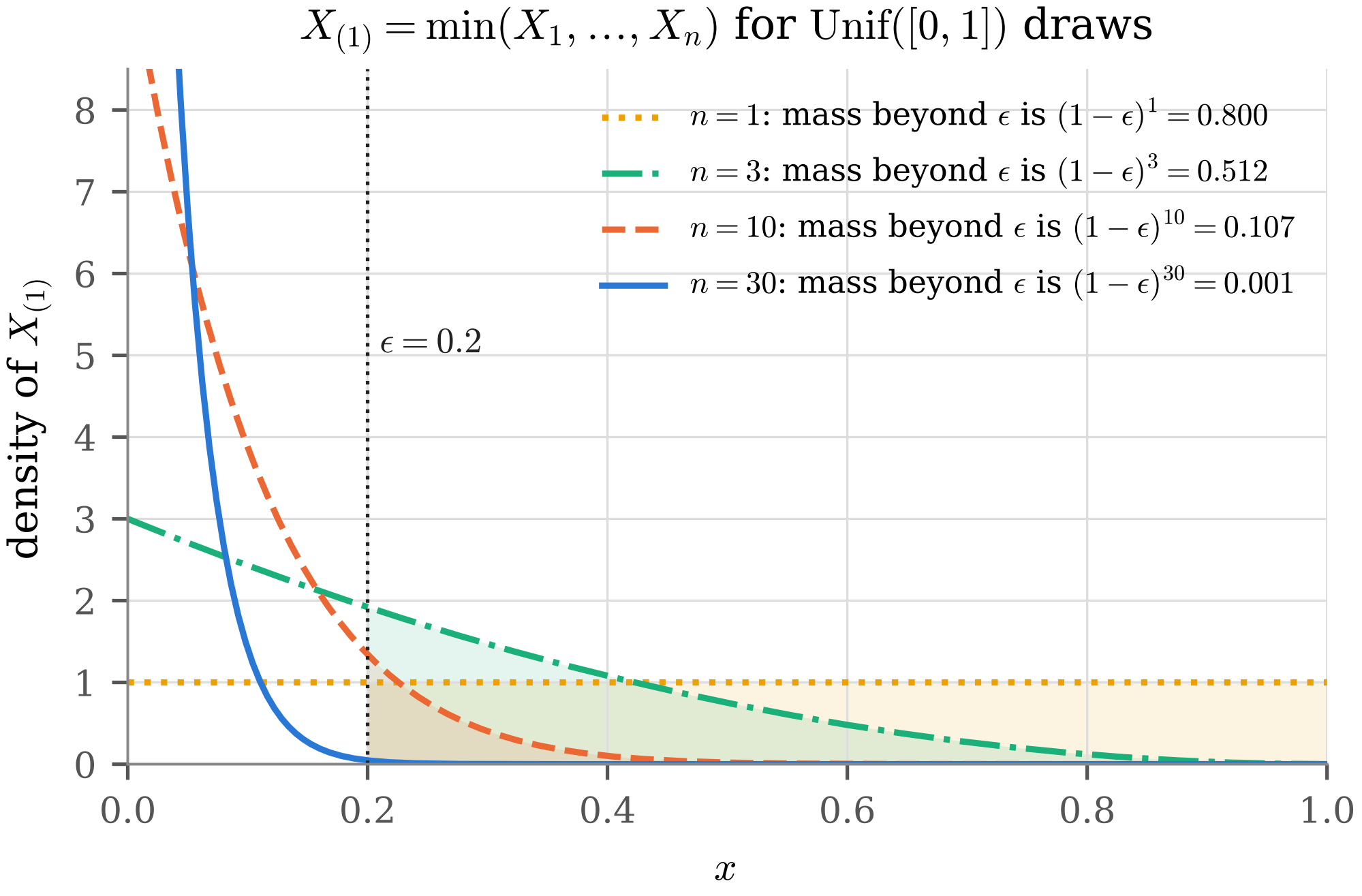
$X_{(1)} = \min(X_1, \ldots, X_n)$ for Unif([0, 1]) draws
$n = 1$: mass beyond ϵ is $(1-\epsilon)^1 = 0.800$
$n = 3$: mass beyond ϵ is $(1-\epsilon)^3 = 0.512$
$n = 10$: mass beyond ϵ is $(1-\epsilon)^{10} = 0.107$
$n = 30$: mass beyond ϵ is $(1-\epsilon)^{30} = 0.001$
$\epsilon = 0.2$
density of $X_{(1)}$
0
1
2
3
4
5
6
7
8
0.0
0.2
0.4
0.6
0.8
1.0
x